OLS Regression - Diagnostic Audit Report

Target: **revenue** | Predictors: **ad_spend** | Best Model: **Linear** | n = **150**

OLS Regression Plot - Best Model (Linear)

Diagnostic Scorecard - Best Model (Linear)

Test	Statistic	p-value	Threshold	Result	Notes
F-Test (Overall)	8828.2957	0.0000	p < 0.05	**PASS**	df model=1, resid=148
Normality - Shapiro-Wilk	0.9831	0.0631	p > 0.05	**PASS**	Tests residual normality
Normality - Jarque-Bera	5.6590	0.0590	p > 0.05	**PASS**	Skew=0.267, Kurt=3.788
Homoscedasticity - Breusch-Pagan	0.0071	0.9327	p > 0.05	**PASS**	Tests constant variance
Autocorrelation - Durbin-Watson	2.1373	-	1.5 – 2.5	**PASS**	Tests residual independence
Multicollinearity - Max VIF	1.00	-	< 5	**PASS**	Predictors: ad_spend
Influential Obs - Cook's D	8 flagged	-	4/n = 0.0267	**FAIL**	Red circles on residual plot

OLS Regression - Interpretation & Model Comparison

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Interpretation of Results

Overall Fit	The best model (degree 1) explains 98.4% of variance in revenue ($R^2 = 0.9835$, Adj $R^2 = 0.9834$), indicating a strong fit.
Model Significance (F-Test)	$F(1, 148) = 8828.296$, $p = 0.0000$. The overall regression is statistically significant ($p < 0.05$).
Normality of Residuals (Shapiro-Wilk)	$W = 0.9831$, $p = 0.0631$. Residuals appear normally distributed — the normality assumption is met.
Homoscedasticity (Breusch-Pagan)	BP stat = 0.0071, $p = 0.9327$. Residual variance appears constant (homoscedastic).
Autocorrelation (Durbin-Watson)	DW = 2.1373. No significant autocorrelation detected.
Influential Observations (Cook's Distance)	8 influential observation(s) detected (Cook's D > 0.0267). Review flagged points for data entry errors or genuine outliers.

Top 8 Models Ranked by BIC

Models ranked by original-scale BIC — lower = better fit penalised for complexity. SW p = Shapiro-Wilk (normality); BP p = Breusch-Pagan (homoscedasticity). Green = PASS (p > 0.05), Red = FAIL (p ≤ 0.05). Single-predictor runs also test Logarithmic, Exponential, Log-Log, Reciprocal, Spline, and Lag-1 variants, producing up to 10 candidates. Multi-predictor runs fit Linear and Interaction models only, so fewer rows appear.

Rank	Model Type	Fitted Equation	R^2	Adj R^2	AIC	BIC	SW p	BP p
1	**Linear**	revenue = -0.430 + 3.2210*ad_spend	0.9835	0.9834	1183.18	1189.2	**0.0631**	**0.9327**
2	**Lag-1**	revenue = -0.487 + 3.2206*ad_spend + 0.0019*ad_spend_lag1	0.9835	0.9833	1178.26	1187.28	**0.0605**	**0.7039**
3	**Spline (Natural Cubic)**	revenue = 130.148 - 123.7901*cr(x, knots=[np.float64(21.478128562638968), np.float64(44.81112389783186), np.float64(74.96618393601378)])[0] - 65.0522*cr(x, knots=[np.float64(21.478128562638968), np.float64(44.81112389783186), np.float64(74.96618393601378)])[1] + 18.7548*cr(x, knots=[np.float64(21.478128562638968), np.float64(44.81112389783186), np.float64(74.96618393601378)])[2] + 105.7684*cr(x, knots=[np.float64(21.478128562638968), np.float64(44.81112389783186), np.float64(74.96618393601378)])[3] + 194.4668*cr(x, knots=[np.float64(21.478128562638968), np.float64(44.81112389783186), np.float64(74.96618393601378)])[4]	0.9847	0.9843	1178.0	1193.06	**0.4132**	**0.5432**
4	**Polynomial (deg 2)**	revenue = 0.403 + 3.1683*ad_spend + 0.0005*ad_spend^2	0.9835	0.9833	1185.02	1194.06	**0.0615**	**0.8871**
5	**Polynomial (deg 4)**	revenue = 6.678 + 1.7428*ad_spend + 0.0706*ad_spend^2 - 0.0012*ad_spend^3 + 0.0000*ad_spend^4	0.9844	0.984	1180.79	1195.84	**0.2878**	**0.6823**
6	**Polynomial (deg 3)**	revenue = -1.323 + 3.3831*ad_spend - 0.0050*ad_spend^2 + 0.0000*ad_spend^3	0.9836	0.9832	1186.54	1198.58	**0.0658**	**0.9169**
7	**Logarithmic**	revenue = -139.867 + 82.8873*ln(ad_spend)	0.76	0.7584	1584.87	1590.89	**0.0**	**0.0**
8	**Reciprocal**	revenue = 164.664 - 185.6768*1/ad_spend	0.1428	0.137	1775.84	1781.86	**0.0**	**0.0008**

Diagnostic Plots - Best Model